ECAE: Edge-Aware Class Activation Enhancement for Semisupervised Remote Sensing Image Semantic Segmentation

Wang Miao, Zhe Xu, Jie Geng, *Member, IEEE*, and Wen Jiang

***Abstract*— Remote sensing image semantic segmentation (RSISS) remains challenging due to the scarcity of labeled data. Semisupervised learning can leverage pseudolabels to enhance the model's ability to learn from unlabeled data. However, accurately generating pseudolabels for RSISS remains a significant challenge that severely affects the model's performance, especially for the edges of different classes. To overcome these issues, we propose a semisupervised semantic segmentation framework for remote sensing images (RSIs) based on edge-aware class activation enhancement (ECAE). First, the baseline network is constructed based on the average teacher model, which separates the training of labeled and unlabeled data using student and teacher networks. Second, considering local continuity and global discreteness of object distribution in RSIs, the class activation mapping enhancement (CAME) network is designed to predict local areas more remarkably. Finally, the edge-aware network (EAN) is proposed to improve the performance of edge segmentation in RSIs. The combination of the CAME with the EAN further heightens the generation of high-confidence pseudolabels. Experiments were performed on two publicly available remote sensing semantic segmentation datasets, Potsdam and ISPRS Vaihingen, which verify the superiorities of the proposed ECAE model.**

***Index Terms*— Class activation mapping, remote sensing images (RSIs), semantic segmentation, semisupervised learning.**

I. Introduction

REMOTE sensing image semantic segmentation (RSISS) is a critical component of remote sensing intelligent interpretation [1], [2], [3], [4], [5], [6], [7], wherein the objective is to assign a semantic label to each pixel in the image [8]. Efficient semantic segmentation techniques are the cornerstone for realizing remote sensing images' (RSIs) intelligent interpretation and downstream task applications [9], [10], [11].

Continuous advancements in deep learning methodologies have unlocked substantial potential for elevating a multitude of applications [12], [13], [14], [15], [16]. For instance, contemporary approaches, such as the multiscale interactive fusion network, graph-feature-enhanced selective assignment network, structural optimization transmission network, and representation-enhanced status replay network, have demonstrated substantial performance enhancements across a diverse array of domains [17], [18], [19], [20], [21]. In the domain of hyperspectral unmixing, Hong et al. [22] introduced an innovative augmented linear mixing model (LMM). This model adeptly addresses spectral variability through a data-driven learning approach tailored to solve hyperspectral unmixing problems. Simultaneously, Li et al. [23] proposed an automatic clustering-based two-branch convolutional neural network (CNN) designed for hyperspectral image classification. Their approach encompasses spectral variance reduction, suppression of spectral amplitude interference, and spatial feature extraction, culminating in precise classification. Furthermore, Hong et al. [24] embarked on a comprehensive exploration of CNNs and graph convolutional networks (GCNs) for hyperspectral image classification. They adeptly tackled the computational challenges associated with the traditional GCNs, particularly in the context of large-scale remote sensing tasks, through the development of a multimodal deep learning (MDL) framework. In addition, they ventured into cross-modality learning (CML) applications, a recurring theme in RSI classification. Nie et al. [25] introduced a novel scale–relation joint decoupling network (SRJDN) tailored for semantic segmentation in RSIs. This network skillfully considers decoupling scales and relationships, thereby facilitating a more comprehensive understanding of multiscale remote sensing objects. Finally, Long et al. [26] unveiled the UIU-Net, a "U-Net in U-Net" framework meticulously crafted for the detection of small objects in infrared images. The UIU-Net proficiently enhances both global and local contrast information through a multilevel and multiscale representation learning approach. These pioneering approaches collectively fuel advancements in various tasks within remote sensing and hyperspectral image analysis.

Thanks to the strong representation capabilities of deep models, significant progress has also been made in applying deep learning methods for high-resolution RSISS tasks [5]. Wu et al. [27] first applied the fully convolutional network (FCN) to automatic building segmentation and significantly enhanced the accuracy of building segmentation using multientropy constraints on different scale feature maps.

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Wu et al. [28] improved the DeepLab v3+ network by proposing a dynamic multiscale context segmentation network, which captures more contextual features by training an extended network with different patch sizes. Song and Civco et al. [13] proposed a road extraction method using a dual U-Net network for joint training. By linking two U-Net networks indirectly, the model's depth was increased to improve the feature-fitting ability of the network. Wang et al. [29] presented a novel semantic segmentation network called the hidden feature-guided network (HFGNet). HFGNet achieves precise semantic segmentation by extracting and fusing valuable feature information hierarchically. Li et al. [4] constructed a synergistic attention module (SAM) that jointly models both spatial and channel affinity, which aims to address the problem of attention bias for RSISS.

One of the major challenges in RSISS is the limited availability of labeled data and the abundance of unlabeled data, leading to a weak supervision scenario [30], [31], [32], [33], [34]. Insufficient training samples can lead to overfitting for a deeper neural network [35], [36], [37], [38]. There have been many recent studies on semisupervised RSISS. To enhance the consistency of model training, a semisupervised segmentation method called paradigms integration and contrastive selection (PICS) is proposed [31]. PICS ensures greater consistency during model training and leads to improved segmentation performance. Wang et al. [39] presented a semisupervised method called RanPaste for RSISS. The method uses pseudolabels generated by randomly integrating fragments of the ground-truth labels into the predicted segmentation map. Li et al. [30] proposed a semisupervised strategy for RSISS, which uses a generative adversarial network (GAN) through self-training. Lu et al. [40] introduced a semisupervised learning approach based on linear sampling self-training, named LSST, for RSISS. The LSST builds upon the traditional pseudolabeling-based paradigm and incorporates strong data augmentations to improve performance. In addition, incorporating the teacher–student model has proven to be instrumental in enhancing the performance of semisupervised segmentation. For instance, Xu et al. [41] introduced an innovative and efficient end-to-end hybrid consistency regularization algorithm (EI-HCR) tailored for semisupervised semantic segmentation of RSI. Their approach optimally leverages a limited number of labeled images alongside a larger set of unlabeled images. By integrating knowledge distillation, they streamlined the student network, reducing model inference time. In addition, Wang et al. [42] proposed a semisupervised segmentation method based on an iterative contrastive network for RSIs. By combining a few labeled images with a larger set of unlabeled images, their method significantly enhances the model's performance. Yang et al. [43] incorporated label propagation and contrastive regularization. Their method augments unlabeled images by randomly copy–paste class regions from labeled images, and a prototype feature constraint module is used to enforce feature alignment across the entire dataset using prototype features from labeled images. Zhang et al. [44] proposed a deep adaptive fusion network with multiple field of views, featuring a multiscale deformable threshold feature extraction module and mixed attention mechanism. Their pyramid ensemble structure, combining deformable convolution and bole convolution, effectively extracts features of objects with varying shapes and sizes while reducing redundant features. The mixed attention mechanism aggregates long-range contextual relationships and fuses low-level and high-level features.

Although these semisupervised segmentation methods achieve outstanding performance, there are drawbacks in generating pseudolabels, especially for the edges of different classes. This article proposes a semisupervised semantic segmentation method for RSIs based on edge-aware class activation enhancement (ECAE). The model consists of three main modules: the pseudolabel self-training network that relies on the average teacher model, the pseudolabel generation network based on class activation mapping enhancement (CAME), and the edge-aware network (EAN) for pseudolabel refinement. The contributions of our work can be summarized as follows.

1) To prevent overfitting on labeled data during training for semisupervised RSISS, we use an average teacher-model-based network to differentiate between labeled and unlabeled samples. It uses a selective edge-aware self-training scheme to ensure the consistency of the model training process by enhancing the quality of the samples.
2) Considering local continuity and global discreteness of object distribution in RSIs, a pseudolabel generation method based on CAME is proposed to improve the confidence of the pseudolabels. It enhances the class activation map locally to make target predictions more accurate in local areas.
3) To address the issue of inaccurate edge segmentation in RSIs, we propose an EAN that leverages image samples and predicted pseudolabels to reduce cumulative errors and achieve more accurate segmentation of edge pixels. Moreover, the combination of CAME and EAN can generate high-quality pseudolabels for semisupervised RSISS.

The subsequent sections of our study are organized as follows. Section II provides a concise overview of the relevant literature. The ECAE method, which we propose, is elaborated in Section III. Section IV presents the comparative results and experimental analyses. Finally, Section V summarizes our conclusions.

II. Related Work

A. Image Semantic Segmentation

Image segmentation is a major challenge in computer vision that has been tackled with numerous methods [45], [46], [47]. These include early techniques such as encoders and decoders, skip-connections, multiscale architectures, and more recent approaches such as the use of dilated convolutions [45]. For instance, Yin et al. [46] endeavored to close the gap between semantic and instance segmentation by introducing Sem2Ins, a real-time model for instance segmentation. The approach effectively generates instance boundaries based on semantic segmentation using conditional GANs (CGANs). Yang et al. [47] introduced a method called part-based semantic transform

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(PST) for aligning object semantics between support and query images. Their approach uses semantic decomposition-and-match, where object semantics are decomposed into multiple prototypes corresponding to object parts using prototype mixture models. The decomposition process is facilitated by an expectation–maximization algorithm. Mehta et al. [48] found that a straightforward encoder–decoder architecture featuring a ResNet-like backbone and a small multiscale head performs equally well or outperforms more complex architectures for semantic segmentation tasks.

B. Semisupervised Semantic Segmentation

Semisupervised semantic segmentation is a technique that focuses on training segmentation networks with a limited number of labeled samples and a substantial amount of unlabeled samples. Currently, the mainstream semisupervised semantic segmentation algorithms mainly include models based on consistency regularization and models based on self-training learning [49], [50], [51], [52].

The consistency-regularization-based models primarily leverage the coherence of segmentation outcomes from unlabeled samples under various perturbations. For example, Ouali et al. [49] proposed a cross-consistency regularized segmentation network called CCT, which constructs two twin segmentation networks with different initializations. It fully uses the implicit information of unlabeled samples and enhances the feature representation of the segmentation network. Lai et al. [50] proposed a context-aware consistency network to maintain the consistency of features with the same labels but different contextual information and addressed the problem of poor generalization of semisupervised models to unseen scenes. Zhong et al. [51] introduced a contrastive learning loss into the consistency regularization model. The proposed semantic segmentation network maintains consistency in the label space between enhanced images and in the feature space between different pixels.

Models based on self-training learning mainly generate pseudolabels for unlabeled samples. For example, Zou et al. [53] first used the self-training model with pseudolabels for semantic segmentation networks. Feng et al. [54] studied the problem of noise for pseudolabels from the perspective of model inconsistency and proposed a dynamic reweighted loss function to mutually train two different models together. Ibrahim et al. [55] introduced an auxiliary dataset with target box labels and used an auxiliary model to generate initial segmentation pseudolabels. Kim et al. [56] further considered the impact of class imbalance on pseudolabels and proposed a pseudolabel distribution alignment refinement method, significantly improving semantic segmentation.

Due to limited samples, the existing semisupervised semantic segmentation methods often struggle to cope with underfitting issues at object boundaries. To overcome these challenges, we present a comprehensive framework based on ECAE. Our approach significantly improves segmentation results without requiring additional annotation effort, especially at object boundaries. We leverage the CAME to enhance class activation maps, resulting in more accurate target predictions within local regions. Furthermore, to tackle inaccurate edge segmentation in RSIs, we introduce an EAN that weights features enriched with semantic boundary information, effectively reducing boundary blur. To alleviate the burden of manual labeling, the EAN leverages image samples and predicted pseudolabels to reduce cumulative errors and achieve precise edge pixel segmentation.

III. Methodology

A. Overview

This article proposes a semisupervised RSISS model with CAME and edge perception, as shown in Fig. 1. The model consists of three main components: the pseudolabel self-training network based on the average teacher model, the pseudolabel generation network based on CAME, and the EAN for pseudolabel refinement. The proposed model performs CAME on the main body region and refinement perception on the edge region of the target, which generates accurate and high-quality pseudolabels from unlabeled samples.

For the task of semisupervised RSISS, the goal is to train a semisupervised model using a small labeled sample set $D_l = \{(x_i^l, y_i^l)\}_{i=1}^{N_l}$ and a larger unlabeled sample set $D_u = \{(x_i^u)\}_{i=1}^{N_u}$, where x_i^l and y_i^l are the labeled sample and its label, respectively, x_i^u is the unlabeled sample, and N_l and N_u are the number of labeled and unlabeled datasets, respectively, generally with $N_l < N_u$.

B. Self-Training Network Based on Average Teacher Model

To prevent overfitting on labeled data, we designed a standard encoder–decoder network for RSISS based on the self-training benchmark network. The network includes a student encoder g_θ^s, a student decoder h_θ^s, a teacher encoder g_ϕ^t, and a teacher decoder h_ϕ^t. The student and teacher encoder–decoder networks have identical structures, with θ and ϕ representing their respective network parameters. The student encoder–decoder network is updated using stochastic gradient descent (SGD) during training progress, while the teacher encoder–decoder network parameters are updated by the exponential moving average (EMA) of the student encoder–decoder network without performing gradient backpropagation ϕ

$$\phi_{t+1} = \alpha\phi_t + (1-\alpha)\theta_t \tag{1}$$

where t is the current epoch of training progress, and $\alpha \in (0, 1)$ is the weight of the transfer parameter.

In addition, we sample the same number of labeled and unlabeled samples in each batch, where the batch size is set to $2B$. However, since the size of the unlabeled sample set N_u is larger than the labeled sample set N_l, we repeat the labeled samples in the training process until all the unlabeled samples are sampled. The labeled samples D_l are only used to train the student encoder–decoder network, and the predicted segmentation results are compared with their corresponding labels to compute the supervised loss

$$L_{\text{sup}} = -\frac{1}{|D_l|}\sum_{(x_i^l, y_i^l)\in D_l}\sum_{j=1}^{H\times W} y_{i,j}^l \log\left(g_\theta^s \circ h_\theta^s\left(x_{i,j}^l\right)\right) \tag{2}$$

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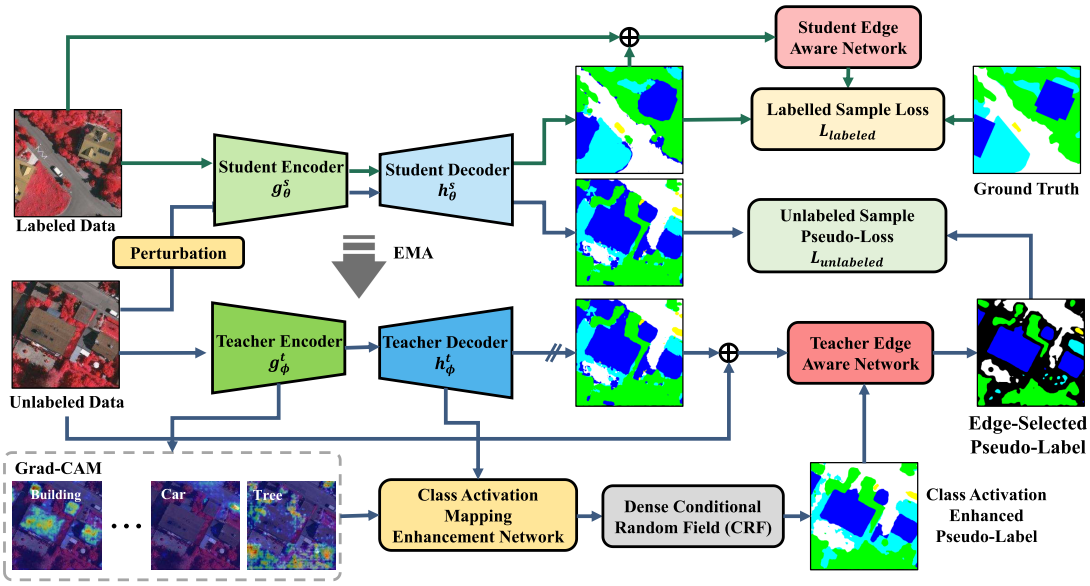


Fig. 1. Overall framework of the proposed ECAE. Our proposed ECAE model combines CAME with EAN for semisupervised RSISS.

where $g \circ h$ denotes the composite function of the codec network. For the unlabeled sample set D_u, after updating the parameters of the teacher codec network, the output prediction $p_i^u = g_\phi^t \circ h_\phi^t(x_i^u)$, taking the index of its maximum value as the pseudolabel of the unlabeled samples $\hat{y}_i^u$ as follows:

$$\hat{y}_i^u = \begin{cases} \text{argmax}\left(p_i^u\right) = \text{argmax}\left(g_\phi^t \circ h_\phi^t\left(x_i^u\right)\right), & \text{if } p_i^u > \tau \\ \text{ignore}, & \text{otherwise} \end{cases} \tag{3}$$

where τ denotes the threshold for pseudolabels, set to 0.7, and the teacher codec network is solely used for generating pseudolabels without performing gradient backpropagation.

To learn the latent features of unlabeled samples, the samples are perturbed and then fed into the student codec network for learning. The pseudolabels generated by the teacher codec network are used to calculate the pseudoloss in conjunction with the unlabeled samples

$$L_{\text{pseudo}} = -\frac{1}{|D_u|} \sum_{x_i^u \in D_u} \sum_{j=1}^{H \times W} \hat{y}_{i,j}^u \log\left(g_\theta^s \circ h_\theta^s\left(f\left(x_{i,j}^u\right)\right)\right) \tag{4}$$

where $f(\cdot)$ refers to the perturbation function, and color jitter and Gaussian blur data augmentation methods are used to perturb the unlabeled samples. The perturbation function essentially uses the idea of consistency regularization, which states that the stochasticity or data augmentation will not change the model's predictions for the same unlabeled sample.

To achieve semisupervised RSISS, we developed a baseline network that leverages an average teacher model and integrates a pseudolabel self-training mechanism. However, the generation of pseudolabel for unlabeled samples is constrained by a fixed threshold, leading the network to focus only on high-confidence pixel features that are easily classified while ignoring a significant number of low-confidence pixel features. As a result, the network tends to overfit simple samples and loses its ability to recognize difficult samples. Furthermore, using a fixed threshold for pseudolabel generation can also result in high-confidence false pseudolabels, negatively affecting the network's performance. Given the characteristics of the target distribution of RSIs, which exhibit local continuity and global discreteness, the proposed method includes a pseudolabel generation network based on CAME to improve the confidence of pseudolabels in the local range. In addition, the EAN is proposed to sense the edge regions of the target in the global range.

C. CAME Network

The target distribution of RSIs is characterized by local continuity and global discreteness. High-density regions in the local range often belong to the same class, and enhancing the feature fusion in these regions can further improve pseudolabels' confidence. Therefore, we propose a pseudolabel generation network based on CAME, as shown in Fig. 2. The network first calculates the weighted gradient-weighted class activation mapping (Grad-CAM) [57] for the feature map of the teacher encoder to generate class activation coarse labels. Then, the class activation coarse labels are jointly inputted with the corresponding features of the teacher decoder into the CAME network. Finally, the enhanced pseudolabels are refined by the dense conditional random field (CRF) [58] to generate a high-confidence pseudolabel.

For the teacher encoder–decoder network, the output feature size of the encoder decreases continuously. In contrast, the output feature size of the decoder increases continuously, finally matching the size of the input image to achieve pixelwise semantic segmentation. Therefore, there are two feature maps with the same size in the output features of the encoder–decoder network, namely, encoder feature $F_e \in \mathbb{R}^{H \times W \times C_e}$ and decoder feature $F_d \in \mathbb{R}^{H \times W \times C_d}$. Encoder feature F_e usually contains more spatial features, while decoder feature F_d contains more semantic features. Therefore, integrating the spatial and semantic information of the encoder

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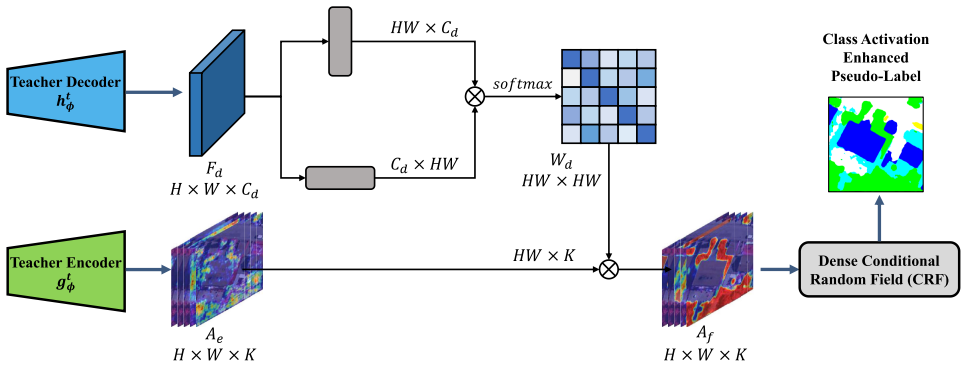


Fig. 2. Progress of CAME. The CAME network generates class activation coarse labels by first calculating the weighted Grad-CAM for the feature map of the teacher encoder. These labels are then fed into the CAME network along with the corresponding features from the teacher decoder.

and decoder features can enhance the confidence of the pseudolabels. Specifically, for the encoder feature F_e, its weighted gradient class activation mapping (Grad-CAM) $A_e \in \mathbb{R}^{H\times W\times K}$ is calculated as follows:

$$A_e = \left\{ \mathrm{ReLU}\left(\sum_{C_e} \alpha_{C_e}^i F_e \right) \right\}_{i=0}^{K} \tag{5}$$

where C_e represents the channel dimension of the encoder feature F_e, K represents the number of classes, ReLU(.) represents the activation function used to remove class activation insensitive regions, and α_e^i represents the activation weight of the ith class, defined as follows:

$$\alpha_e^i = \frac{1}{H \times W} \sum_{j=1}^{H\times W} \frac{\partial \left(g_\phi^t(x^u)\right)^i}{\partial F_j^e} \tag{6}$$

where $(g_\phi^t(x^u))^i$ represents the predicted score of the ith class by the teacher encoder. Grad-CAM uses the gradient information backpropagated to the encoder feature F_e to compute the importance level of each channel C_e, and then weights and sums the encoder feature α_e^i to obtain the class activation map F_e. However, it is not practical to directly use the class activation map A_e as the pseudolabel, as its results are often coarse and not fine-grained. Although Grad-CAM can reflect the distribution characteristics of RSI targets, it is not accurate to generate pseudolabels.

The CAME network enhances each category's class activation maps and generates more refined pseudolabels. The network structure framework is shown in Fig. 2. For the decoder feature F_d extracted by the teacher decoder, compared with the encoder feature F_e, it has more abundant semantic information. According to the local continuity of the object in RSIs, the similarity weights W_d of the local contextual features are calculated based on the self-attention mechanism

$$W_d = \mathrm{softmax}\left(\delta(F_d)^T \delta(F_d)\right) \tag{7}$$

where $\delta(.)$ is the scale transformation function. The feature similarity weights W_d can integrate the contextual information of the encoder feature F_e through self-attention calculation. Then, after performing matrix multiplication with the decoder class activation map A_e and passing through dense CRFs, the enhanced class activation map $A_f \in \mathbb{R}^{H\times W\times K}$ can be obtained

$$A_f = \mathrm{crf}(A_e \otimes W_d). \tag{8}$$

D. Edge-Aware Network

Low-density areas in the global range, often the edge regions of the target, are susceptible to generating high-confidence erroneous pseudolabels. Using these labels as correct pseudolabels for self-training can accumulate errors and significantly impact network performance. To overcome this issue, we propose an EAN that perceives the target's edge regions globally and eliminates high-confidence erroneous pseudolabels. Fig. 3 shows the framework of the EAN.

The proposed EAN selects edges based on predicted pseudolabels and enhanced class activation mapping to obtain the final pseudolabels. The encoder extracts features from input images and their predicted pseudolabels using the ResNet-34 network. The edge-aware feature fusion layer integrates global edge information with different receptive fields, using three depth-separable convolutions with varying dilation rates. The decoder uses multiple 3×3 convolutional and upsampling layers to upsample the fused features. The binary fully connected classifier generates the edge mask and identifies high-confidence incorrect pseudolabels. The EAN consists of a student and a teacher network with identical structures. The student network is trained on labeled samples, and the teacher network uses unlabeled samples to generate edge-selection pseudolabels, meeting the training requirements of the average teacher model.

For the labeled sample x_i^l, its predicted output result by the student encoder–decoder network is $p_i^l = g_\theta^s \circ h_\theta^s(x_{h,w}^l)$, and the pixelwise pseudocross-entropy E_i^l is calculated as follows:

$$E_i^l = \sum_{h=1}^{H} \sum_{w=1}^{W} p_{i,h,w}^l \log\left(p_{i,h,w}^l\right) \tag{9}$$

where h and w refer to the (h, w)th pixel, and $E_i^l \in \mathbb{R}^{H\times W\times 1}$ reflects the reliability of the predicted result $p_i^l \in \mathbb{R}^{H\times W\times K}$. Concatenating the label samples x_i^l, predicted results p_i^l, and their calculated pseudocross-entropy E_i^l along the channel dimension yields $B_i^l \in \mathbb{R}^{H\times W\times(d_0+K+1)}$, where d_0 is the number of input channels for the RSIs, and K is the number of classes. By inputting B_i^l into the student EAN, the edge mask $M_i^l \in [0, 1]^{H\times W}$ can be obtained

$$M_i^l = \mathrm{EAN}\left(B_i^l\right) = \mathrm{EAN}\left(x_i^l \oplus p_i^l \oplus E_i^l\right). \tag{10}$$

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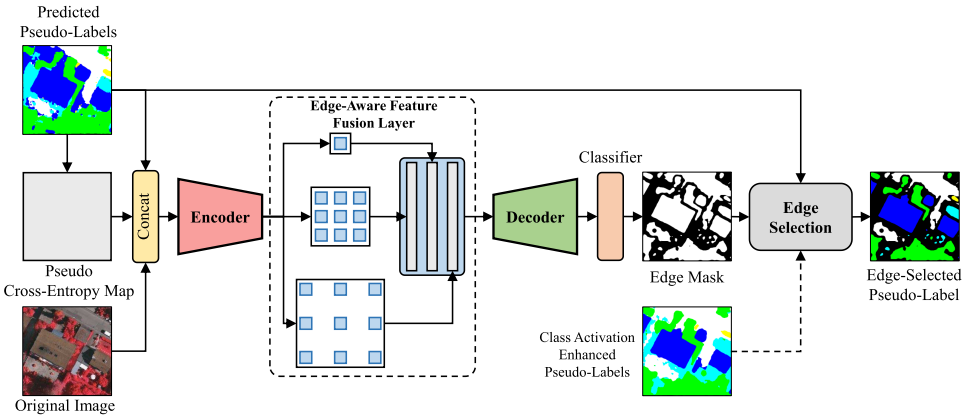


Fig. 3. Framework of EAN. The proposed EAN can perceive target edge regions globally and remove high-confidence erroneous pseudolabels.

The edge mask M_i^l reflects the reliability of the predicted pseudolabel and can locate high-confidence erroneous pseudolabels. In the supervised training process, its target mask M_i^{gt} is defined as follows:

$$M_i^{gt} = \begin{cases} 1, & \text{if } \arg\max\left(p_i^l\right) = y_i^l \\ 0, & \text{otherwise .} \end{cases} \tag{11}$$

The edge-aware supervision loss is given by

$$L_{EAN} = -\frac{1}{|D_l|} \sum_{\left(x_i^l, y_i^l\right) \in D_l} \sum_{j=1}^{H \times W} M_{i,j}^{gt} \log\left(M_{i,j}^l\right). \tag{12}$$

The teacher EAN only predicts edge masks without training for unlabeled samplesx_i^u. Following the above steps, we can also obtain the edge mask M_i^u of the unlabeled sample and perform edge selection with the predicted result $p_i^u = g_\phi^t \circ h_\phi^t(x_i^u)$ obtained by the teacher encoder–decoder network, which leads to the modification of equation as follows:

$$\hat{y}_i^u = \begin{cases} \arg\max\left(p_i^u\right), & \text{if } p_i^u > M_i^u \\ \text{ignore}, & \text{otherwise .} \end{cases} \tag{13}$$

To perform edge selection, we jointly use the edge mask M_i^u, the predicted result p_i^u of the teacher encoder–decoder network, and the class activation enhanced pseudolabel p_i^{cam} if available. The formula is then modified as follows:

$$\hat{y}_i^u = \begin{cases} \arg\max\left(p_i^{cam}\right), & \text{if } p_i^{cam} > p_i^u > M_i^u \\ \arg\max\left(p_i^u\right), & \text{else } \left(p_i^u > p_i^{cam}\right) \ \& \ \left(p_i^u > M_i^u\right) \\ \text{ignore}, & otherwise. \end{cases} \tag{14}$$

The edge mask serves as a confidence threshold, enabling the differentiation between correct and incorrect pseudolabels. When the probability distribution of the predicted or enhanced pseudolabels surpasses the mask threshold, it indicates the presence of correct pseudolabels. Higher probability distribution values signify more reliable and high-confidence correct pseudolabels in such cases. Conversely, when the probability distributions of the predicted or enhanced pseudolabels fall below the mask threshold, both the pseudolabels are identified as incorrect. They should be excluded during the training process. Based on Fig. 4, when p_i^{cam} has the highest confidence, the corresponding prediction at that pixel location, denoted as p_i^{cam}, is used as the high-confidence correct pseudolabel. Similarly, when p_i^u has the highest confidence, the prediction p_i^u at that pixel location is used as the high-confidence correct pseudolabel. This approach ensures the selection of reliable pseudolabels and improves the overall accuracy (OA) and reliability of the segmentation process.

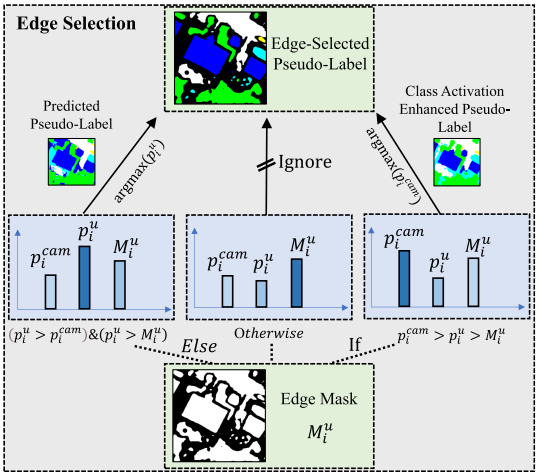


Fig. 4. Framework of EAN. The proposed EAN can perceive target edge regions globally and remove high-confidence erroneous pseudolabels.

The EAN generates selected edge pseudolabels that can globally perceive edge features and remove incorrect pseudolabels in edge regions. Unlike the fixed threshold used by the self-training pseudolabel benchmark network, dynamic filtering based on edge masks can further remove high-confidence incorrect pseudolabels to prevent serious cumulative errors during the self-training phase that can affect network performance.

For the proposed model, the computation consists of four parts, namely, the computation of the baseline encoder, the baseline decoder, the CAME, and the EAN, denoted as $\Omega(BE)$, $\Omega(BD)$, $\Omega(\text{CAME})$, and $\Omega(\text{EAN})$, respectively. Specifically, for a feature map, including the number of input channels C_i, output channels C_o, height H, and width W, the computational complexity $\Omega(\text{ECAE})$ of the proposed model can be written as below

$$\begin{aligned} \Omega(\text{ECAE}) &= \Omega(BE) + \Omega(BD) + \Omega(\text{CAME}) + \Omega(\text{EAN}) \\ &= \left[2C_i HWC_o{}^2 N_{BE}\right] + \left[2C_i HWC_o{}^2 N_{BD}\right] \end{aligned}$$

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TABLE I
NUMBER OF LABELED SAMPLES WITH DIFFERENT SETTINGS FOR THE POTSDAM AND VAIHINGEN TRAINING SETS

Dataset	1/16	1/8	1/4	1
Potsdam	628	1256	2513	10051
Vaihingen	46	92	183	732

$$+\left[2HWC_i{}^2\right]+\left[2C_iHWC_o{}^2N_{\mathrm{EAN}}\right] = 2(N_{BE}+N_{BD}+N_{\mathrm{EAN}})C_iHWC_o{}^2+2HWC_i{}^2 \tag{15}$$

where N_{BE} represents the number of baseline encoder layers, N_{BD} denotes the count of baseline decoder layers, and N_{EAN} corresponds to the number of EAN layers.

IV. EXPERIMENTS

A. Datasets and Experimental Setting

To assess the efficacy of the proposed semisupervised semantic segmentation approach for RSIs, experiments are performed on the Potsdam and Vaihingen datasets. These datasets consist of six typical land cover categories: Surface, Building, Low-Vegetation, Tree, Car, and Background. The Background is excluded from the evaluation because it contains all other miscellaneous objects and has a small sample size. The Potsdam dataset comprises 38 high-resolution RSIs measuring 6000×6000 in size, while the Vaihingen dataset consists of 33 high-resolution RSIs with dimensions approximately 2500×2000. Due to memory limitations, the input images are cropped into 512×512 image patches with a 256-pixel step during the training phase and into 512×512 image patches without overlap during the validation and test phases.

To meet the requirements of the semisupervised learning experiments, the dataset partitioning setting in this section is adjusted. The test and validation sets remain unchanged, and the training set is divided into labeled and unlabeled sample sets. The $1/4$, $1/8$, and $1/16$ samples are randomly selected from the training set as the labeled dataset, and the remaining samples are used as the unlabeled dataset. The specific settings of the Potsdam and Vaihingen training sets are shown in Table I. We evaluate the performance of our proposed method for RSIs by comparing it with the existing state-of-the-art methods on two datasets, namely, Potsdam and Vaihingen. The methods for comparison include the naive-student benchmark network [59], s4GAN [52], CCT [49], and CST [30].

B. Parameter Setup

In this section, we use ResNet-101 as the encoder and DeepLabv3+ as the decoder in the baseline network [60], [61]. The CAME network selects the penultimate layer feature of the teacher encoder as the encoder feature Fe and the second layer feature of the student encoder as the decoder feature Fd. ResNet-34 is used as the edge-aware coding network for the EAN. The specific network structures are detailed in Table II. The student encoder is trained with the SGD optimizer using a weight decay of 0.0001, impulse of 0.9, initial learning rate of 0.005, and decay rate of 0.9. We set the moving average coefficient for updating the teacher encoder parameters to 0.99 and use color jitter and Gaussian blur for data enhancement of unlabeled samples. The training epochs are set to 50. We evaluate the proposed method using three metrics: intersection over union (IoU), F1 score, and OA. We compute metrics including mean IoU (mIoU), mean F1 score (mF1), OA for the entire test set, and F1 scores for individual classes.

C. Results on the Potsdam Dataset

The comparison results of different methods on the Potsdam dataset, under the condition that the number of labeled samples accounts for $1/16$ of the training set, are shown in Table III, where the best-performing result is highlighted in bold. To better illustrate the performance difference between the semisupervised and supervised methods, we have also included the performance results of supervised and fully supervised methods in Table III, where the supervised method refers to the results obtained using only $1/16$ labeled samples, and the fully supervised method refers to the results obtained using all the samples.

From Table III, compared with other mainstream semisupervised RSISS methods, the proposed method achieved state-of-the-art performance improvements in mIoU, mF1, and OA, reaching 75.57%, 85.84%, and 88.01%, respectively. Specifically, our method exhibits notable improvements of 2.53%, 2.74%, and 0.47% in the "Surface," "Building," and "Tree" categories compared with the CCT, and 1.19%, 0.82%, 3.02%, 1.44%, and 4.82% improvement, respectively, compared with o the baseline network. These performance gains are attributed to the effectiveness of our approach in enhancing edge confidence within local high-density regions using the CAME network and EAN. Consequently, the overall segmentation performance experiences significant enhancement. However, for the "Low-Vegetation," "Tree," and "Car" categories in the Potsdam dataset, their edges are relatively well-defined, resulting in more limited performance improvement by our model.

Compared with the supervised and fully supervised method, the proposed model improved 4.19%, 2.95%, and 2.58% in mIoU, mF1, and OA, respectively, using only a fraction of labeled samples in the training set, reaching 95.55%, 96.88%, and 97.83% of the performance of the fully supervised methods in mIoU, mF1, and OA, respectively, demonstrating that the proposed method can fully use a large number of unlabeled samples and achieve performance levels comparable to the fully supervised methods.

In addition, the proposed method achieved the best F1 performance in the "Surface," "Building," and "Tree" categories, outperforming all other methods. For these categories, the target distribution presents local continuity and global discreteness. The proposed method leverages the CAME network to enhance pseudolabel confidence in local high-density areas and the edge-awareness network to remove high-confidence false labels in target edge areas globally. The proposed approach effectively improves pseudolabel confidence and enhances segmentation performance. However, the "Low-Vegetation" and "Car" categories present strong similarities, resulting in a similar target distribution density that makes it challenging for the edge awareness network to distinguish

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TABLE II
INFORMATION ABOUT THE PROPOSED MODEL

	Model	The Number of Layers	Para	FLOPs(G)
Baseline Encoder	ResNet-101	101	44.55M	7.83
Baseline Decoder	DeepLabv3+	50	25.56M	4.11
EAN	Resnet34	34	21.80M	3.67

TABLE III
RESULTS ON THE POTSDAM DATASET UNDER THE CONDITION THAT THE NUMBER OF LABEL SAMPLES IS 1/16 OF THE TRAINING SET

Method	$F1(\%)$					mIoU	mF1	OA
	Surface	Building	Low-Vegetation	Tree	Car	(%)	(%)	(%)
Supervised Method	87.73	92.71	72.91	78.50	82.62	71.38	82.89	85.43
Baseline Network [59]	88.94	94.28	74.96	79.61	80.10	72.43	83.58	86.77
s4GAN [52]	87.49	91.65	80.48	80.37	83.26	74.23	84.65	85.37
CCT [49]	87.60	92.36	**80.48**	80.58	**88.04**	75.44	85.81	85.01
Our Method	**90.13**	**95.10**	77.98	**81.05**	84.92	**75.57**	**85.84**	**88.01**
Fully-Supervised Method	91.52	96.61	83.07	82.67	89.14	79.09	88.60	89.96

between them. Consequently, the "Low-Vegetation" category's accuracy is slightly lower. Nevertheless, the proposed method still achieved the second-best segmentation performance, demonstrating its rationality and effectiveness.

Tables IV and V show the comparison results of various methods on the Potsdam dataset with 1/8 and 1/4 label sample numbers of the training set. These results are consistent with those obtained with 1/16 label sample numbers. The proposed method, which uses CAME and edge perception, achieved the best segmentation performance based on overall mIoU, mF1, and OA evaluation indicators. Specifically, when the number of label samples accounted for 1/4 of the training set, the proposed method achieved 99.48%, 99.49%, and 99.87% in terms of mIoU, mF1, and OA evaluation indicators, respectively, which is comparable to the performance of the fully supervised methods. According to Tables IV and V, we observe a similar trend for the categories "Low-Vegetation" and "Tree" as seen in Table III. The reason behind this trend is that these categories, namely, "Low-Vegetation" and "Tree," do not heavily rely on edge information for accurate segmentation. As a result, their performance might not fully reflect the capabilities of our model. In addition, the strong similarities between the "Low-Vegetation" and "Tree" categories lead to a similar target distribution density, posing a challenge for the edge-awareness network to distinguish between them. Consequently, the accuracy of both the "Low-Vegetation" and "Tree" categories is slightly lower. On the other hand, our model continues to demonstrate robust performance for categories such as "Surface" and "Building," which exhibit more complex edge conditions. These categories benefit from the ECAE, allowing our model to maintain high accuracy. These results demonstrate that the proposed method can effectively use the weak information from a large number of unlabeled samples to generate high-confidence pseudolabels, which leads to superior performance in semisupervised RSISS.

Fig. 5 shows the visualization results of different methods on the Potsdam dataset, where the number of labeled samples in the training set is 1/16. We selected five representative visualizations from the test set to illustrate the superiority of the proposed method. The proposed method based on CAME and edge awareness effectively segments most land objects,

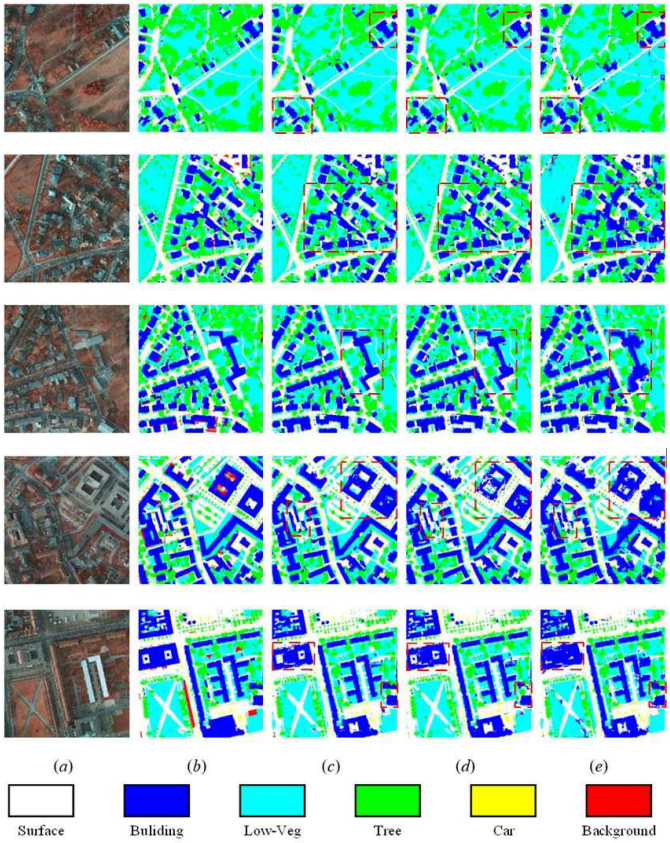


Fig. 5. Visualization results of different methods on the Potsdam dataset. (a) IRRG. (b) GT. (c) ECAE. (d) Baseline. (e) CCT.

except for the background clutter category, when compared with the ground truth. To quantitatively assess the impact of our framework on boundary improvement, we compared it with CCT, specifically focusing on the performance at object boundaries. As depicted in Fig. 5, the experimental results demonstrate the superior performance of our proposed framework in enhancing segmentation at the boundaries. We emphasize the visual comparison of segmentation results, particularly for categories such as "Surface" and "Building," which involve more complex edge situations. Our framework achieves a notable improvement in segmentation accuracy in these cases. On the other hand, the output maps generated by

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TABLE IV
RESULTS ON THE POTSDAM DATASET UNDER THE CONDITION THAT THE NUMBER OF LABEL SAMPLES IS 1/8 OF THE TRAINING SET

Method	$F1(\%)$					mIoU	mF1	OA
	Surface	Building	Low-Vegetation	Tree	Car	(%)	(%)	(%)
Supervised Method	72.01	64.00	39.05	68.70	5.06	36.19	49.76	61.95
Baseline Network [59]	90.29	95.83	79.06	81.83	79.13	74.88	85.23	88.63
s4GAN [52]	88.64	92.01	81.36	81.93	84.09	75.62	85.61	86.49
CCT [49]	87.31	91.12	80.32	81.62	79.36	72.60	83.95	85.22
CST [30]	89.18	92.59	**82.32**	**83.39**	86.63	76.92	86.82	87.07
Our Method	**91.33**	**95.93**	79.76	82.00	**87.41**	**77.94**	**87.29**	**89.51**
Fully-Supervised Method	91.52	96.61	83.07	82.67	89.14	79.09	88.60	89.96

TABLE V
RESULTS ON THE POTSDAM DATASET UNDER THE CONDITION THAT THE NUMBER OF LABELED SAMPLES IS 1/4 OF THE TRAINING SET

Method	$F1(\%)$					mIoU	mF1	OA
	Surface	Building	Low-Vegetation	Tree	Car	(%)	(%)	(%)
Supervised Method	72.01	64.00	39.05	68.70	5.06	36.19	49.76	61.95
Baseline Network [59]	90.85	95.59	78.84	82.25	87.47	77.49	87.00	89.09
s4GAN [52]	88.81	92.92	81.20	80.78	85.09	76.62	85.76	87.42
CCT [49]	88.33	93.00	81.26	**83.04**	82.34	75.09	85.59	86.46
CST [30]	89.67	93.83	**82.75**	82.22	86.92	77.39	87.08	87.26
Our Method	**91.58**	**96.32**	81.93	82.53	**88.39**	**78.68**	**88.15**	**89.84**
Fully-Supervised Method	91.52	96.61	83.07	82.67	89.14	79.09	88.60	89.96

TABLE VI
RESULTS ON THE VAIHINGEN DATASET UNDER THE CONDITION THAT THE NUMBER OF LABEL SAMPLES IS 1/16 OF THE TRAINING SET

Method	$F1(\%)$					mIoU	mF1	OA
	Surface	Building	Low-Vegetation	Tree	Car	(%)	(%)	(%)
Supervised Method	72.01	64.00	39.05	68.70	5.06	36.19	49.76	61.95
Baseline Network [59]	81.96	85.56	60.56	80.76	25.90	54.05	66.95	77.85
s4GAN [52]	82.53	84.02	72.36	82.88	58.49	61.92	76.06	79.56
CCT [49]	81.93	85.34	70.97	81.80	63.66	62.94	76.74	80.51
Our Method	**85.51**	**90.19**	**72.77**	**84.11**	**74.52**	**69.20**	**81.42**	**83.35**
Fully-Supervised Method	86.97	92.12	74.49	84.46	78.19	71.79	83.25	84.66

CCT tend to exhibit certain shortcomings, with some class boundaries being lost and a lack of complete visual regularity. This discrepancy further emphasizes the advantage of our proposed framework in effectively preserving and refining object boundaries. Although the CCT method can roughly recognize land objects, its results appear rougher at the edges of the objects. In contrast, the proposed method has better edge refinement, which further demonstrates the effectiveness of the proposed edge-awareness network.

D. Results on the Vaihingen Dataset

The Vaihingen dataset has lower spatial resolution and fewer samples than the Potsdam dataset. With the same semisupervised training set proportion, the number of labeled samples in the Potsdam dataset is over ten times more than that in the Vaihingen dataset, making the semisupervised RSISS more challenging.

Table VI presents the comparison results of various methods when the number of labeled samples accounts for only 1/16 of the training set on the Vaihingen dataset. From the results in Table III, the proposed method achieved the highest mIoU, mF1, and OA of 69.20%, 81.42%, and 83.35%, respectively, on the Vaihingen test set with only 46 labeled samples. These scores are 6.26%, 4.68%, and 2.84% higher than the second-best CST method. Thus, the proposed semisupervised method outperforms all other methods, proving its superiority. When only 46 labeled samples were used, the mIoU of supervised method was as low as 36.19%, and the car category could barely be detected. This suggests that overfitting can occur when the sample size is too small, leading to poor network performance. However, compared with the supervised method, the proposed method improved the mIoU, mF1, and OA by 33.01%, 31.66%, and 21.40%, respectively. This highlights the proposed method's ability to effectively leverage the weak information of many unsupervised samples, resulting in significant improvements in network performance and robustness. Moreover, the proposed method achieves 96.39%, 97.80%, and 98.45% of the performance of the fully supervised network in mIoU, mF1, and OA, respectively. This demonstrates that the proposed method effectively achieves performance levels comparable to the fully supervised network. Moreover, our proposed method outperforms all other methods in various categories, including "Surface," "Building," "Low-Vegetation," "Tree," and "Car." Specifically, compared with the second-best method, our approach achieves significant improvements of 3.58%, 4.85%, 1.80%, 2.31%, and 10.86% in these respective categories. It is worth noting that the Vaihingen and Potsdam datasets have different sample resolutions, resulting in scale differences between the "Building" and "Car" categories. The "Car" category poses a more complex edge situation, making it challenging for the model to recognize accurately. However, our model effectively captures the edges of these categories through the EAN, leading to superior results compared with other models. The proposed method demonstrates its ability to learn and exploit correlation relationships between features from different resolutions and scales, which in turn enhances the discriminative nature of edge features. This enhancement is particularly evident in

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TABLE VII
RESULTS ON THE VAIHINGEN DATASET UNDER THE CONDITION THAT THE NUMBER OF LABEL SAMPLES IS 1/8 OF THE TRAINING SET

Method	$F1(\%)$					mIoU	mF1	OA
	Surface	Building	Low-Vegetation	Tree	Car	(%)	(%)	(%)
Supervised Method	72.01	64.00	39.05	68.70	5.06	36.19	49.76	61.95
Baseline Network [59]	82.47	85.83	57.28	79.55	34.25	54.44	67.88	77.19
s4GAN [52]	83.09	84.97	73.31	83.06	60.28	63.40	76.94	81.16
CCT [49]	81.30	84.66	69.64	81.72	40.71	57.99	71.61	79.54
CST [30]	83.03	86.33	**73.57**	83.72	63.24	64.68	77.98	81.82
Our Method	**86.50**	**91.22**	72.94	**84.30**	**76.48**	**70.45**	**82.29**	**84.04**
Fully-Supervised Method	86.97	92.12	74.49	84.46	78.19	71.79	83.25	84.66

the "Building" and "Car" categories, further validating the effectiveness of our approach.

Tables VII and VIII show the comparison results of different methods on the Vaihingen dataset with 1/8 and 1/4 label samples. The results are consistent with those under 1/16 label samples, where the proposed semantic segmentation method achieved the best segmentation performance in terms of mIoU, mF1, and OA for each category. Our proposed method uses CAME and edge awareness to generate high-confidence pseudolabels for unlabeled samples from the perspective of local target saliency and edge refinement. The proposed approach significantly improves the network's segmentation performance.

Fig. 6 shows the visualization results of different methods on the Vaihingen dataset under the condition of 1/16 labeled samples in the training set. To highlight the superiority of our proposed method, we selected five representative test set visualization results. In Fig. 6, the images present a wider road with a higher density of vehicles, making it easier to distinguish buildings from the road. Moreover, the shaded area of the buildings appears larger. However, some challenges are evident as well. The grass, buildings, and road are in close proximity, and the road's color resembles that of some buildings, leading to confusion. Specifically, the fine classification of the edges of the largest building is incomplete, and the edge classification of the shaded area is inadequate. Furthermore, the road cannot be effectively distinguished from buildings with similar colors, resulting in unsatisfactory fine classification outcomes. Subsequently, in the analysis, Fig. 6 demonstrates relatively accurate edge classification results for the "Car" category, but improvement is still required for the "Low-Vegetation" and "Tree" categories. Similarly, distinguishing between the road and buildings with similar colors remains challenging. However, the edge classification of large buildings sees significant improvement, reducing misclassifications caused by building shadows. In contrast, our model is superior in distinguishing between buildings and roads, particularly in ambiguous regions. As illustrated in Fig. 6, our proposed method for RSIs can accurately and finely segment various targets, such as buildings with different shapes and large-scale differences, trees embedded in low vegetation and complex edges, artificial ground with significant intraclass differences often covered by shadows, and cars with discrete target distribution and small feature sizes. Particularly, for the building category, the baseline method and the CCT method produce rough segmentation results, which are prone to misclassification and inaccurate edge recognition. In contrast, our proposed method achieves accurate edge detection and complete segmentation results of targets, even with a small number of labeled samples. These results further demonstrate the effectiveness of our proposed method.

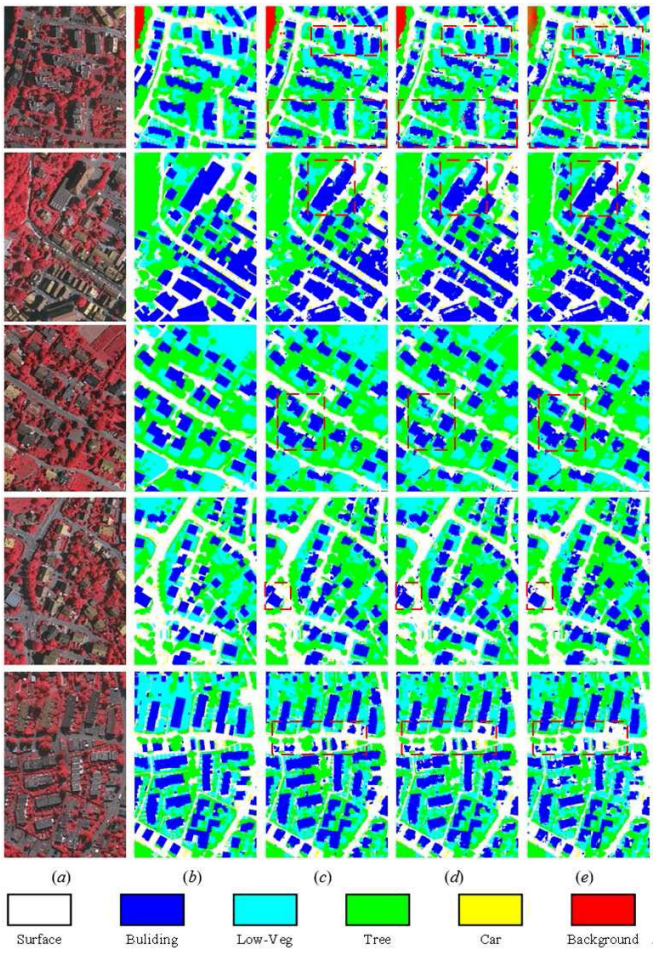


Fig. 6. Visualization results of different methods on the Vaihingen dataset. (a) IRRG. (b) GT. (c) ECAE. (d) Baseline. (e) CCT.

E. Ablation Experiment

To further validate the effectiveness of the proposed method, the ablation experiments were conducted on the Vaihingen dataset to analyze the performance of the proposed CAME network and edge awareness network. Since the Vaihingen dataset has fewer samples than the Potsdam dataset, the ablation experiments conducted on the Vaihingen dataset better demonstrate the semisupervised performance of the proposed method under extremely few labeled samples.

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TABLE VIII

RESULTS ON THE VAIHINGEN DATASET UNDER THE CONDITION THAT THE NUMBER OF LABEL SAMPLES IS 1/4 OF THE TRAINING SET

Method	$F1$(%)					mIoU	mF1	OA
	Surface	Building	Low-Vegetation	Tree	Car	(%)	(%)	(%)
Supervised Method	72.01	64.00	39.05	68.70	5.06	36.19	49.76	61.95
Baseline Network [59]	83.33	88.54	70.56	81.29	60.94	63.53	76.93	77.85
s4GAN [52]	85.44	88.78	75.61	82.91	66.86	67.26	79.92	83.30
CCT [49]	85.16	89.84	**76.75**	82.25	54.14	64.99	77.63	83.54
CST [30]	85.67	88.96	76.11	83.48	68.69	68.29	80.58	83.68
Our Method	**86.42**	**91.64**	74.89	**84.99**	**76.95**	**71.39**	**82.98**	**84.64**
Fully-Supervised Method	86.97	92.12	74.49	84.46	78.19	71.79	83.25	84.66

TABLE IX

ABLATION EXPERIMENTS ON THE VAIHINGEN DATASET UNDER THE CONDITION THAT THE NUMBER OF LABELED SAMPLES IS 1/16 OF THE TRAINING SET

Method	$F1$(%)					mIoU	mF1	OA
	Surface	Building	Low-Vegetation	Tree	Car	(%)	(%)	(%)
Baseline Network [59]	81.96	85.56	60.56	80.76	25.90	54.05	66.95	77.85
Baseline Network + CAME	83.21	88.52	69.11	81.32	66.72	64.40	77.78	80.72
Baseline Network + EAN	85.23	90.02	70.00	82.82	72.8	67.57	80.17	82.47
Baseline Network + CAME + EAN	**85.51**	**90.19**	**72.77**	**84.11**	**74.52**	**69.20**	**81.42**	**83.35**

TABLE X

ABLATION EXPERIMENTS ON THE VAIHINGEN DATASET UNDER THE CONDITION THAT THE NUMBER OF LABELED SAMPLES IS 1/8 OF THE TRAINING SET

Method	$F1$(%)					mIoU	mF1	OA
	Surface	Building	Low-Vegetation	Tree	Car	(%)	(%)	(%)
Baseline Network [59]	82.47	85.83	57.28	79.55	34.25	54.44	67.88	77.19
Baseline Network + CAME	85.51	89.57	70.48	82.14	61.74	64.91	77.89	82.16
Baseline Network + EAN	85.10	89.89	71.20	83.33	70.26	67.31	79.96	82.69
Baseline Network + CAME + EAN	**86.50**	**91.22**	**72.94**	**84.30**	**76.48**	**70.45**	**82.29**	**84.04**

TABLE XI

ABLATION EXPERIMENTS ON THE VAIHINGEN DATASET UNDER THE CONDITION THAT THE NUMBER OF LABELED SAMPLES IS 1/4 OF THE TRAINING SET

Method	$F1$(%)					mIoU	mF1	OA
	Surface	Building	Low-Vegetation	Tree	Car	(%)	(%)	(%)
Baseline Network [59]	83.33	88.54	70.56	81.29	60.94	63.53	76.93	77.85
Baseline Network + CAME	84.18	89.55	72.07	82.79	65.21	65.82	78.76	82.33
Baseline Network + EAN	85.70	91.02	73.46	84.31	71.87	69.10	81.27	83.87
Baseline Network + CAME + EAN	**86.42**	**91.64**	**74.89**	**84.99**	**76.95**	**71.39**	**82.98**	**84.64**

Tables IX–XI present the performance of the CAME network and EAN on the Vaihingen dataset, with labeled sample ratios of 1/16, 1/8, and 1/4. Incorporating the CAME network into the baseline network at a labeled sample ratio of 1/16 results in substantial improvements in the mIoU, mF1, and OA metrics by 10.35%, 10.83%, and 2.87%, respectively. The CAME network effectively fuses encoder and decoder features using self-attention based on the weighted gradient class activation map, leading to an improved class activation coarse label. This enhancement generates high-confidence pseudolabels, significantly boosting the network's segmentation performance. Particularly in the "Car" category, where weak and insufficient edge information poses challenges, the feature enhancement notably improves the model's segmentation performance. Furthermore, Tables IX–XI demonstrate notable performance improvements after incorporating the EAN into the baseline network. At a labeled sample ratio of 1/16, the mIoU, mF1, and OA metrics improve by 13.52%, 13.22%, and 4.62%, respectively. This enhancement is particularly significant as the baseline network suffers from performance degradation due to accumulated errors. The EAN effectively perceives the global target edge regions and can remove erroneous pseudolabels with high confidence, thereby effectively mitigating the impact of accumulated errors on the network's performance. By integrating the CAME network and EAN into the baseline network, the proposed network achieves the best performance on the Potsdam and Vaihingen datasets, even under different labeled sample size conditions. Tables IX–XI indicate that the CAME network and EAN complement each other, enhancing the main area of the target object in the pseudolabels and providing precise perception of the edge area of the target object. This combination results in accurate and high-quality pseudolabels generated from unlabeled samples, ultimately leading to excellent segmentation performance under semisupervised conditions. Furthermore, leveraging the feature enhancement and using EAN to optimize the high-confidence pseudolabels of edges effectively improves the segmentation performance for categories with weak information and edges.

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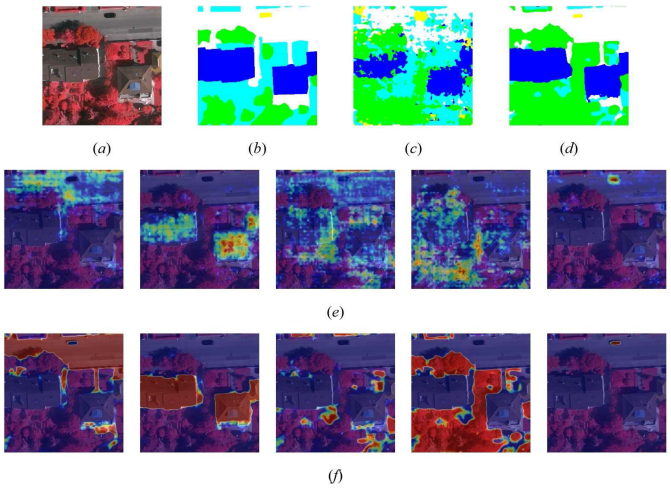


Fig. 7. Visualization results with the CAME network. (a) RGB. (b) GT. (c) Pseudolabels. (d) Enhanced pseudolabels. (e) Class activation mapping for each class. (f) Enhanced class activation mapping for each class.

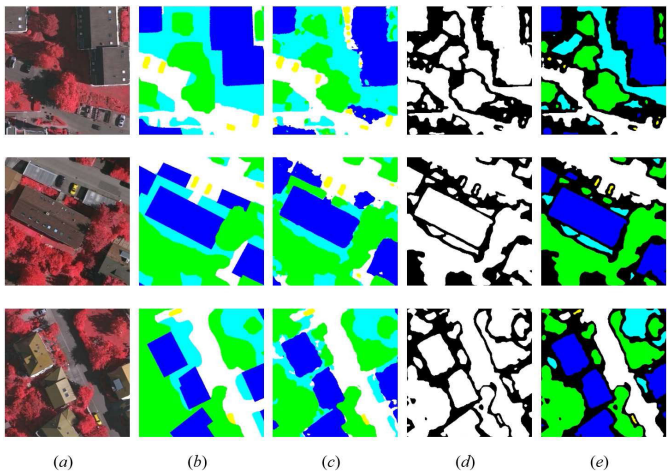


Fig. 8. Visualization results with the EAN. (a) RGB. (b) GT. (c) Pseudolabel. (d) Edge mask. (e) Edge selection results.

The visualization results of applying the CAME network to the Vaihingen dataset are shown in Fig. 7. As seen in Fig. 7, the direct application of the weighted gradient class activation map results in rough and imprecise pseudolabels. However, the pseudolabels become more completed and refined after enhancing the coarse class activation label using the proposed CAME network. The proposed approach is capable of generating more accurate pseudolabels, with the heatmap's color depth in each class activation mapping reflecting the network's confidence in that pixel. Furthermore, the confidence of the target region is observed to be higher after applying the CAME network. Thus, the proposed network can enhance high-density regions in the local area and generate high-confidence pseudolabels.

Fig. 8 shows the visualization results with the EAN to the Vaihingen dataset. According to Fig. 8, while the predicted pseudolabels generated by the teacher decoder can effectively segment the main body of the targets, there are still issues with edge detection and prediction errors in low-density regions where two classes intersect. The proposed EAN addresses these issues by perceiving edge regions globally, removing false pseudolabels with high confidence, and allowing

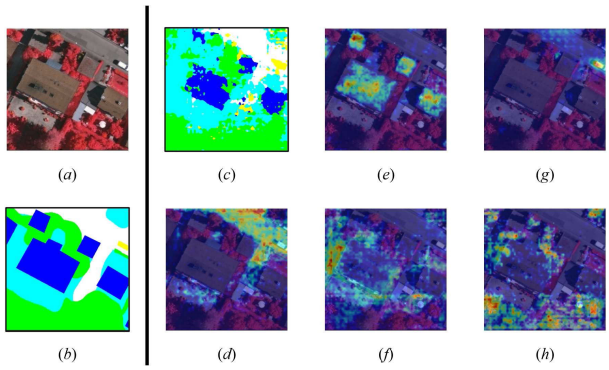


Fig. 9. Results of various types of Grad-CAM on the Vaihingen dataset and their pseudolabels. (a) Original image. (b) GT. (c) CAM pseudolabel. (d) Surface. (e) Building. (f) Low-vegetation. (g) Car. (h) Tree.

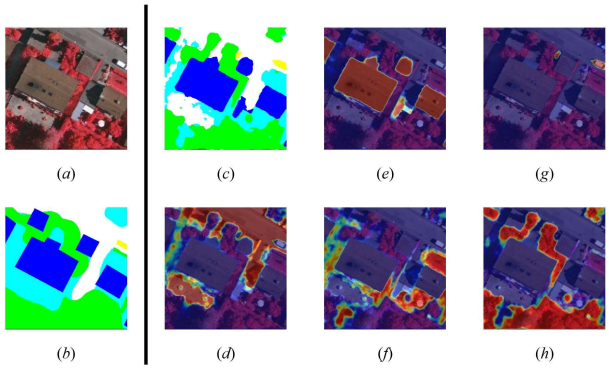


Fig. 10. Class activation enhancement maps for each class generated on the Vaihingen dataset and their pseudolabels. (a) Original image. (b) GT. (c) CAM pseudolabel. (d) Surface. (e) Building. (f) Low-vegetation. (g) Car. (h) Tree.

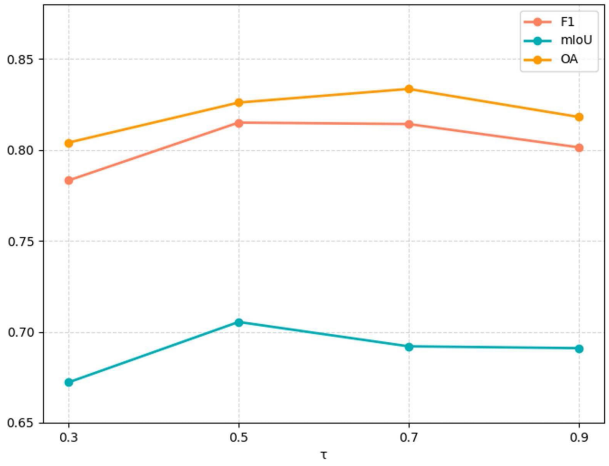


Fig. 11. Sensitivity analysis of parameter τ.

uncertain regions to relearn to avoid cumulative errors during self-training. The visualization results provide further evidence of the effectiveness of the network.

Figs. 9 and 10 display the enhanced class activation maps and their corresponding pseudolabels for each class on the Vaihingen dataset. The maps exhibit higher confidence in each target, resulting in more accurate segmentation, and the generated pseudolabels are more refined. These refined labels can be used to facilitate self-training learning of the baseline network.

Fig. 11 presents the test results with sensitivity analysis of parameter τ under 1/16 samples on the Vaihingen dataset. The

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analysis of the results reveals that extreme values of the parameter τ adversely affect the model's segmentation accuracy. The observed phenomenon can be attributed to the following reasons: When τ is set to a very small value, the selected pseudolabeled samples tend to be inaccurate, consequently impacting the model's overall performance. Conversely, when τ is chosen to be excessively large, the scarcity of positive samples reduces the model's accuracy. Striking a suitable balance for τ is critical to achieving optimal segmentation results.

V. Conclusion

This article proposes a semisupervised method for RSISS based on ECAE. The proposed method achieves outstanding segmentation performance on the Potsdam and Vaihingen datasets. Our experiments demonstrate that the CAME network enhances the class activation map locally, resulting in more accurate target predictions in local areas. Furthermore, the EAN can perceive the target's edge regions globally and eliminate high-confidence erroneous pseudolabels. High-quality pseudolabels are generated by combining the class activation enhancement and EANs, greatly improving segmentation performance.

In future work, we will continue to focus on enhancing edge perception in remote sensing segmentation tasks. Furthermore, we intend to further enhance the robustness of our proposed method to handle a wider range of variabilities and degradations commonly encountered in remote sensing data. This includes investigating techniques to better address spectral variability, noise, and atmospheric effects.

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